

Auditing Multi-Task Graph Neural Networks for Political Tweet Classification

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Abstract—We compare graph convolution, neighborhood sampling, and attention for two tasks on a collection of 7,053 political tweets: identifying one of 14 authors and predicting a binary misinformation pseudo-label. An initial graph construction placed a direct edge between every tweet and its author account, making author identification partly an edge lookup. We remove those edges and account-profile features before evaluating three two-layer multi-task graph neural networks (GNNs) on a fixed tweet split. Across three initialization seeds, author macro-F1 ranges from 0.820 to 0.865 by architecture. GraphSAGE obtains 0.702 misinformation macro-F1, comparable to a text-only logistic regression baseline at 0.703. The audit changes the interpretation of the experiment: the graph supports author prediction, but these results do not establish an advantage over text alone for pseudo-label prediction.

Index Terms—graph neural networks, leakage audit, political tweets, weak supervision, misinformation

I. Introduction

Graph neural networks combine a node’s features with information from its neighbors. This is attractive for social-media analysis, where a tweet’s text and the accounts that engage with it may both carry useful signals. It also creates an evaluation hazard: if a target is encoded as an edge or feature, impressive classification scores need not represent generalization.

This study asks whether three standard GNN encoders—GCN [1], GraphSAGE [2], and GAT [3]—can jointly predict a tweet’s author and an automatically assigned misinformation label. We first audit the graph representation, then compare the corrected GNNs with simple non-graph baselines on the same split. The principal contribution is the corrected comparison and its interpretation, rather than a claim of reliable misinformation detection.

II. Data and Labels

The dataset contains 7,053 tweets from 14 Macedonian political accounts. The initial heterogeneous graph has 7,053 tweet nodes, 3,107 account nodes, and 97,364 directed edges. Tweet features consist of 256 TF-IDF text dimensions and five standardized numeric metadata fields. Account nodes have zero-valued features. Edges connect tweets to their authors and to accounts listed as retweeters; reverse edges permit bidirectional message passing.

The second target is derived from a multilingual zero-shot classifier, joeddav/xlm-roberta-large-xnli, using the candidate labels valid, misleading, and invalid. The observed counts are 5,303, 666, and 1,084, respectively. We merge the latter two into a positive binary class. These are

model-generated labels, not independently verified fact-checks. Thus the experiment tests agreement with a label generator, not factual truth.

III. Leakage Audit and Experimental Design

In the original graph, each tweet has a direct link to its author account. Since author identity is the first target, this is target leakage through topology. Exactly 14,106 directed edges—one in each direction for each tweet—are removed. An explicit post-filter check finds zero remaining direct tweet–author edges. We also zero the two tweet feature dimensions holding the author’s follower and following counts, which could act as account fingerprints. Retweeter relations remain. The corrected graph has 83,258 directed edges.

Tweets are stratified by author into 4,513 training, 1,129 validation, and 1,411 test examples using a fixed split (seed 42). The graph is transductive: all nodes and non-target edges are visible, but only training tweet labels enter the loss. GNN input TF-IDF features were fitted on the full unlabeled tweet collection in the original preparation workflow; the text-only baseline fits its vocabulary on training tweets. This difference is a limitation of the comparison.

Each encoder has two message-passing layers (96 hidden units), layer normalization, ReLU, dropout, and separate task heads. The summed cross-entropy loss weights the positive pseudo-label class by two. AdamW uses a learning rate of 0.008, weight decay of 10^{-4} , and at most 35 epochs. Checkpoints are selected by mean post-update validation macro-F1 across tasks, with early stopping after eight non-improving epochs. Initializations use seeds 42, 43, and 44 on the fixed split.

For the pseudo-label task, we compare against majority-class prediction and class-balanced logistic regression on word unigram/bigram TF-IDF features (up to 30,000 terms). The vectorizer is fitted on the training tweets only. Accuracy and macro-F1 are computed on the common test set; macro-F1 matters because the classes are imbalanced. GNN scores are means $\pm$ sample standard deviations across three initialization seeds; baselines use one deterministic fit.

IV. Results

Tables I and II report the two tasks separately. GAT has the highest mean author macro-F1 after the audit. GraphSAGE has the highest mean GNN pseudo-label macro-F1, but its 0.702 is effectively tied with text-only logistic regression’s 0.703. The small standard deviations describe initialization variability on one fixed split, not uncertainty across datasets or future events.

TABLE I
Author identification after leakage removal (test macro-F1).

GNN encoder	Author macro-F1
GCN	0.820 ± 0.007
GraphSAGE	0.857 ± 0.016
GAT	0.865 ± 0.003

TABLE II
Binary pseudo-label prediction (test macro-F1).

Model	Pseudo-label macro-F1
Majority class	0.428
Text logistic regression	0.703
GCN	0.688 ± 0.003
GraphSAGE	0.702 ± 0.009
GAT	0.669 ± 0.004

The original, unaudited graph produced author macro-F1 values of 0.957 (GCN), 0.989 (GraphSAGE), and 0.997 (GAT). Those figures are retained only as a leakage diagnostic and are not valid evidence of author-identification performance. For seed 42 on the corrected graph, GraphSAGE obtains author accuracy 0.858 and pseudo-label accuracy 0.771, with macro-F1 scores 0.842 and 0.708. Its pseudo-label confusion matrix, with rows denoting true negative/positive, is $\begin{bmatrix} 871 & 184 \\ 139 & 217 \end{bmatrix}$. Text logistic regression obtains accuracy 0.755 and confusion matrix $\begin{bmatrix} 827 & 228 \\ 118 & 238 \end{bmatrix}$. The graph model makes fewer false-positive predictions but more false negatives on this split.

V. Discussion and Limitations

Removing authorship edges substantially lowers author scores while leaving a meaningful but imperfect signal in text and retweeter context. The corrected result is a useful demonstration of how graph construction determines what a GNN can legitimately learn. It does not isolate the benefit of topology from that of text and metadata, nor does it prove that joint training is superior to a single-task model on the corrected graph.

The pseudo-label task has a second interpretation problem: the target was itself inferred from tweet text. A text-only model can learn to imitate that labeling procedure without detecting misinformation. The small difference between GraphSAGE and logistic regression offers no compelling evidence that the graph adds value for this target. Further limits include one random, author-stratified split,

possible overlap of related tweets across partitions, TF-IDF fitted on all unlabeled text for the GNN, and the lack of external fact-check annotations. A stronger next study would use independently verified labels, temporal or account-held-out evaluation, and graph-versus-no-graph ablations under identical feature fitting.

VI. Conclusion

This project compares three multi-task GNNs after auditing and removing a direct authorship shortcut. GAT is strongest for corrected author identification, while GraphSAGE is strongest among GNNs for the generated misinformation label. The latter does not outperform a simple text-only baseline in macro-F1. Careful graph construction and honest baselines are therefore as important as architecture choice in this setting.

Code Availability

Source code and the Colab workflow are available in the GitHub repository.

References

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